

(TR-104) PYTHON WITH AI/ML

Training Day 77 Report

Date: 28 April 2026

Topic Covered: Trailer Analytics, AI Tool Enhancement, Agent Optimization & Integration Testing

The seventy-seventh day of training focused on enhancing trailer analytics capabilities, improving AI tool selection logic, optimizing agent responses, and performing integration testing across backend and frontend components.

- Added trailer-level analytics tools for identifying leasing customers and analyzing pending payments.
- Developed additional tools for customer insights, insurance lookup, fleet financial summaries, and trailer-wise revenue analysis.
- Integrated customer search functionality using identifiers such as DL numbers, SSN, and policy numbers.
- Implemented direct insurance policy lookup and fleet financial reporting tools.
- Improved AI tool selection logic to better align responses with user intent and analytical query requirements.
- Enhanced decision-making to prioritize summary-based responses over unnecessary detailed listings.
- Updated the chat API to include original user queries in responses for improved traceability and debugging.
- Simplified and optimized agent system prompts to improve response quality and formatting consistency.
- Conducted extensive testing of payment, trailer, insurance, and revenue-related queries.
- Validated query accuracy, tool selection behavior, and response formatting across multiple scenarios.
- Tested Ollama integration through Ngrok and verified remote model connectivity using LangChain.
- Performed frontend-backend WebSocket integration testing and validated real-time communication flow.

- Revised concepts related to AI tool design, analytics generation, prompt engineering, intent-based tool selection, and remote LLM integration.

Overall, the training enhanced understanding of AI-driven analytics, tool-based architectures, prompt optimization, integration testing, and intelligent query handling systems.